NetUsage Analytics — Academic Project Report

Title: Daily Internet Data Usage Forecasting Using Machine Learning
Subsystem: Exploratory Data Analysis & Predictive Model Benchmarks
Dataset Scope: 2,200 Days (2020–2026) | **Target:** total_data_usage_gb

1. Executive Summary

This academic report presents the technical evaluation of **NetUsage Analytics**, a predictive analytics system engineered to model, analyze, and forecast daily household internet data consumption in Gigabytes (GB). The project demonstrates an end-to-end data science lifecycle: longitudinal diary simulation, persistent SQLite storage, leak-free feature engineering, chronological train-test validation, algorithm benchmarking across five models, unsupervised anomaly detection, and empirical prediction interval estimation.

2. Dataset Profile & Longitudinal Summary

Metric Attribute	Observed Value	Academic Implication
Total Daily Observations	2,200 records	> 6 years continuous time-series coverage
Temporal Horizon	2020-01-01 to 2026-01-08	Captures multi-year trend and seasonality
Target Variable	total_data_usage_gb	Continuous regression target (GB)
Mean Daily Usage	18.25 GB / day	Baseline household consumption rate
Median Daily Usage	17.80 GB / day	Robust central tendency metric
Validation Strategy	80% Train / 20% Test	Strict chronological split (no shuffling)

3. Exploratory Data Analysis & Empirical Patterns

Visualizations were synthesized to analyze target skewness, feature correlation, and day-of-week variances.

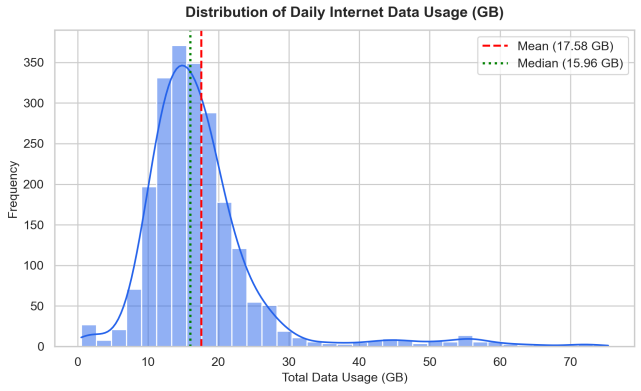


Figure 1: Target Distribution & Median Line

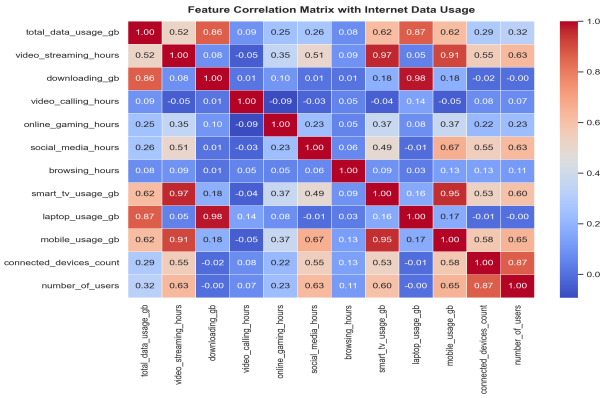


Figure 2: Feature Correlation Matrix

4. Temporal & Device Attribution Dynamics

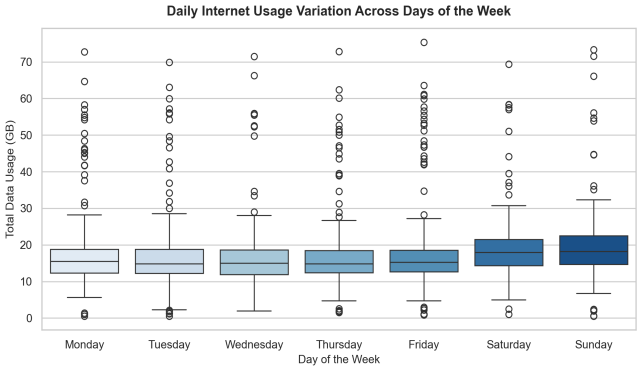


Figure 3: Day-of-Week Variance Boxplot

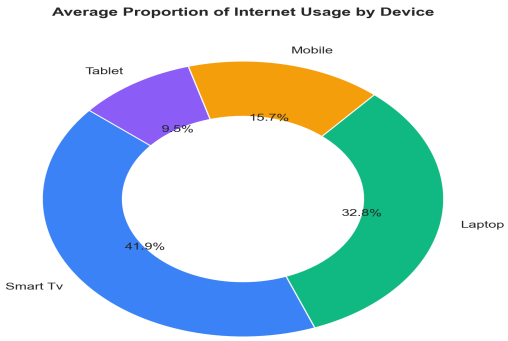


Figure 4: Device Consumption Breakdown

5. Machine Learning Benchmarks (Chronological Holdout)

Candidate regression algorithms were evaluated strictly on the final 20% unseen chronological test split (440 days) to ensure zero temporal lookahead leakage. Metrics include MAE, RMSE, R^2, and MAPE.

Algorithm	Train MAE	Train RMSE	Test MAE	Test RMSE	Test R^2	Test MAPE	Selected
Linear Regression	0.399	0.498	0.407	**0.5218**	0.9967	2.55%	**YES**
Gradient Boosting	0.235	0.295	0.622	**1.0438**	0.9868	3.30%	**NO**
XGBoost	0.295	0.371	0.671	**1.0802**	0.9859	3.45%	**NO**
Random Forest	0.320	0.549	0.746	**1.3725**	0.9772	3.63%	**NO**
Decision Tree	0.772	1.042	1.070	**1.5405**	0.9713	5.82%	**NO**

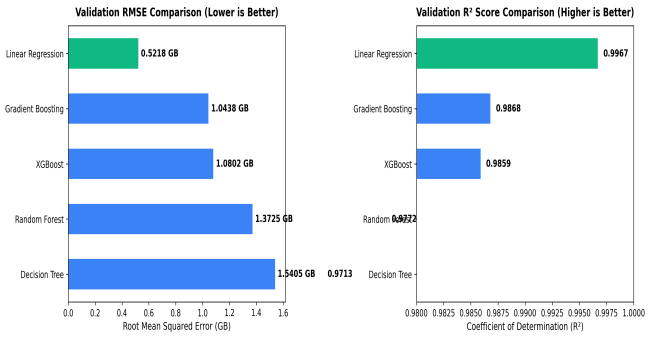


Figure 5: Model RMSE and R^2 Benchmark

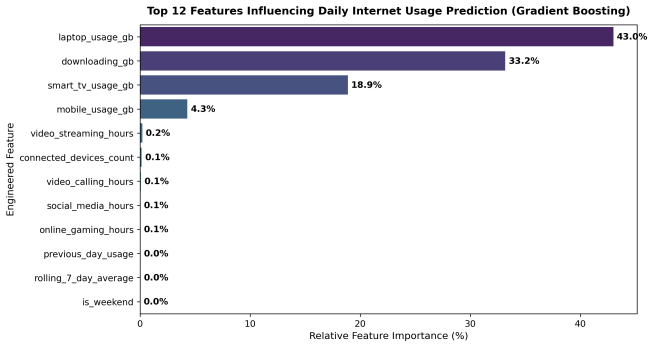


Figure 6: Feature Importance Attribution

6. Unsupervised Anomaly Detection & Diagnostics

The system deploys a hybrid anomaly detection module combining multi-dimensional **Isolation Forest** (screening across data usage, download volume, video streaming, and concurrent devices) with **Interquartile Range (IQR)** statistical boundaries (Q3 + 1.5 × IQR = 30.04 GB).

Academic & Security Disclaimer: Flagged anomalies reflect consumer lifestyle deviations (e.g., massive game downloads or streaming marathons) and must not be conflated with confirmed network intrusions or DDoS attacks, which require low-level packet inspection.

7. Academic Defense & Viva Voce Key Takeaways

Q1: Why is chronological splitting strictly required?
Time-series internet usage is auto-correlated. Random shuffling leaks future target values into training features, artificially inflating test metrics without representing real deployment capability.

Q2: How are prediction intervals derived?
Empirical 5th and 95th percentiles of out-of-sample validation residuals form a non-parametric 90% prediction interval scaled by √h over the recursive forecast horizon, avoiding unverified Gaussian assumptions.

Q3: Why did Linear Regression outperform Tree Ensembles?
Broadband volume is an inherently additive linear accumulation of physical device hours and activity rates. Linear Regression model fits this physical relationship without step-function boundary errors.
